

SOFTWARE REQUIREMENTS SPECIFICATION FOR EXPENSE TRACKER SYSTEM

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for the **Expense Tracker System**, a software application designed to help users track, manage, and analyze their personal or business expenses efficiently.

1.2 Document Conventions

- Headings are numbered hierarchically.
- “Shall” indicates mandatory requirements.
- “Should” indicates recommended requirements.
- “May” indicates optional features.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Developers
- Project managers
- Stakeholders
- QA engineers

Readers should begin with the Introduction and Overall Description before reviewing detailed features.

. 4 Product Scope

The Expense Tracker allows users to:

- Record daily expenses
- Categorize spending
- View reports and summaries
- Set budgets and financial goals

1.5 References

- IEEE 830 SRS Standard
- Software Engineering Best Practices
- Financial tracking application guidelines

2. Overall Description

2.1 Product Perspective

The Expense Tracker is a standalone application that may integrate with banking APIs or cloud storage systems.

2.2 Product Functions

- Add/Edit/Delete expenses
- Categorize expenses
- Generate reports (daily, weekly, monthly)
- Budget tracking and alerts
- Data backup and synchronization

2.3 User Classes and Characteristics

- **General Users:** Basic knowledge of mobile or web apps
- **Advanced Users:** Require analytics and reporting
- **Admin (optional):** Manages system settings

2.4 Operating Environment

- Mobile: Android / iOS
- Web: Modern browsers (Chrome, Edge, Safari)
- Backend: Cloud-based server (optional)

2.5 Design and Implementation Constraints

- Must support secure authentication
- Must comply with data privacy regulations
- Limited offline functionality depending on sync design

2.6 User Documentation

- User manual
- FAQ section
- In-app help/tutorial

2.7 Assumptions and Dependencies

- Users have internet access for syncing
- Integration APIs (if used) remain available
- Device storage is sufficient

3. External Interface Requirements

3.1 User Interfaces

- Dashboard with summary charts
- Expense entry form
- Category management screen
- Reports and analytics page

3.2 Hardware Interfaces

- Smartphone or computer
- Optional: biometric authentication (fingerprint/face recognition)

3.3 Software Interfaces

- Database (SQL/NoSQL)
- Optional bank APIs
- Cloud storage services

3.4 Communications Interfaces

- HTTPS for secure data transmission
 - REST API for backend communication
-

4. System Features

4.1 Expense Management

- Users can add, edit, and delete expenses
- Each expense includes an amount, category, date, and notes

4.2 Category Management

- Users can create custom categories
- Predefined categories are available

4.3 Reporting and Analytics

- Generate charts and summaries
- Filter by date, category, or amount

4.4 Budget Management

- Set monthly or category-based budgets

Receive alerts when limits are exceeded

4.5 Data Synchronization

- Sync data across devices
- Backup and restore functionality

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system should load within 2 seconds
- Support at least 10,000 transactions per user

5.2 Safety Requirements

- Prevent data loss during crashes
- Regular automatic backups

5.3 Security Requirements

- User authentication (email/password, OAuth)
- Data encryption (in transit and at rest)
- Secure session handling

5.4 Software Quality Attributes

- **Usability:** Simple and intuitive UI
- **Reliability:** Minimal downtime
- **Scalability:** Support increasing users
- **Maintainability:** Modular code structure

5.5 Business Rules

- Expenses must have a valid amount
- Dates cannot be in invalid formats
- Budget alerts triggered when threshold exceeded

6. Other Requirements

- Multi-language support (optional)
- Dark mode UI (optional)

- Export data (CSV/PDF)

Appendix A: Glossary

- **Expense:** Money spent by the user
- **Category:** Classification of expenses (e.g., food, travel)
- **Budget:** Spending limit set by the user

Appendix B: Analysis Models

B.1 Use Case Diagram

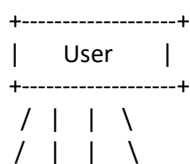
Actors:

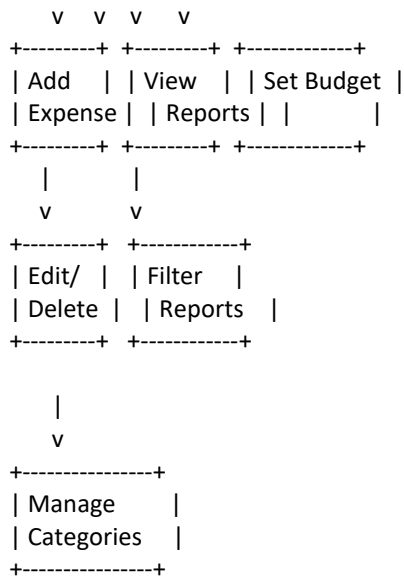
- User
- Admin (optional)

Use Cases:

- Register/Login
- Add Expense
- Edit/Delete Expense
- View Reports
- Set Budget
- Manage Categories
- Sync Data

Use Case Diagram (Textual Representation)





B.2 Detailed Use Case Description

Use Case: Add Expense

- **Actor:** User
- **Description:** User adds a new expense entry
- **Precondition:** User is logged in
- **Main Flow:**
 1. User opens expense form
 2. Enters amount, category, date
 3. Clicks save
 4. System stores expense
- **Post condition:** Expense is saved successfully

Use Case: View Reports

- **Actor:** User
- **Description:** User views expense summaries
- **Main Flow:**
 1. User selects report type
 2. System retrieves data
 3. Displays charts/graphs

B.3 Sequence Diagrams

Sequence Diagram 1: Add Expense

User UI Backend Database

| | | |

| --Enter-> | | |

| Data | | |

| | --Send----> | |

| | Request | |

| | | --Insert---> | |

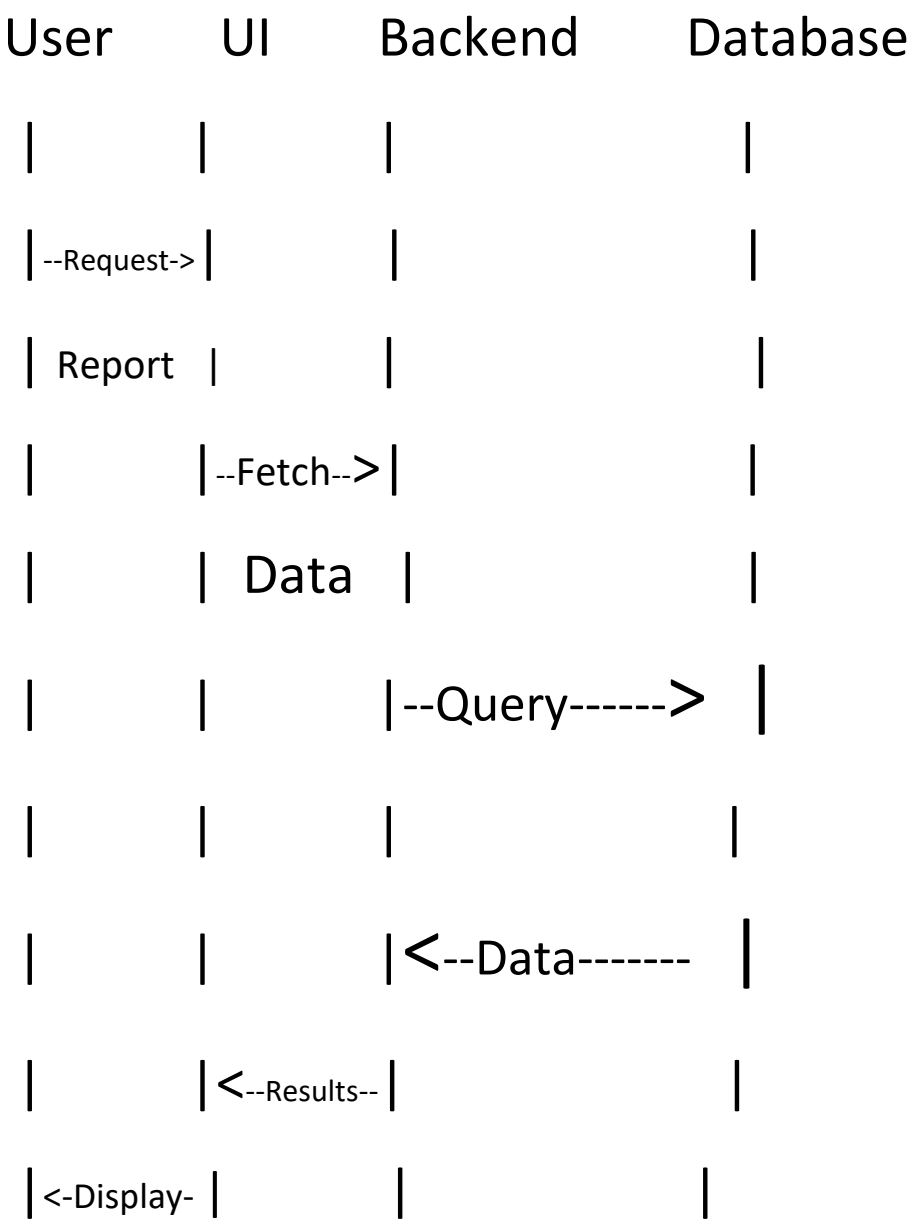
| | | Record | |

| | | <--Success-- | |

| | <--Response- | |

| <--Confirm- | | |

Sequence Diagram 2: View Reports



Appendix C: To Be Determined List

- Integration with specific banking API s
- Advanced AI-based spending insights
- Multi-user collaboration features

